Exercise 1. We say a set $F \subset \mathbb{R}$ is closed if its complement $F^c := \mathbb{R} \setminus F$ is open (see Assignment 3 for a discussion of open sets). Since $\emptyset$ and $\mathbb{R}$ are open, it follows that $\emptyset$ and $\mathbb{R}$ are closed as well.

(a) Let $a, b \in \mathbb{R}$ with $a < b$. Prove that $[a, b]$ is closed.

(b) Is the set of integers $\mathbb{Z} \subset \mathbb{R}$ closed? Provide a proof to substantiate your claim.

(c) Is the set of rationals $\mathbb{Q} \subset \mathbb{R}$ closed? Provide a proof to substantiate your claim.

(d) Let Λ be a set (not necessarily a subset of $\mathbb{R}$), and for each $\lambda \in \Lambda$, let $F_\lambda \subset \mathbb{R}$. Prove that if F_λ is closed for all $\lambda \in \Lambda$ then the set

$$\bigcap_{\lambda \in \Lambda} F_\lambda = \{x \in \mathbb{R} : x \in F_\lambda \text{ for all } \lambda \in \Lambda\}$$

is closed.

(e) Let $n \in \mathbb{N}$, and let $F_1, ...F_n \subset \mathbb{R}$. Prove that if $F_1, ..., F_n$ are closed then the set

$$\bigcup_{m=1}^{n} F_m = \{x \in \mathbb{R} : x \in F_m \text{ for some } m = 1, ..., n\}$$

is closed.

Proof. Start!

(a) Let $a, b \in \mathbb{R}$ and $a < b$. By Assignment 3 Exercise 5, $[a, b]^c = (-\infty, a) \cup (a, \infty)$ is open. Thus, $[a, b]$ is closed, as claimed.

(b) Yes, the subset of integers $\mathbb{Z}$ is closed in the set of reals $\mathbb{R}$. We need to prove that $\mathbb{R} \setminus \mathbb{Z}$ is open. Let $x \in \mathbb{R} \setminus \mathbb{Z}$ and $E = \{n \in \mathbb{Z} : n < x\} \subset \mathbb{R}$. Then E is nonempty and bounded above. By the Least Upper Bound Property, $\lfloor x \rfloor := \sup E \in \mathbb{R}$ and $\lceil x \rceil := \sup E + 1 \in \mathbb{R}$. We want to prove that $\lfloor x \rfloor \in \mathbb{Z}$ and $\lceil x \rceil \in \mathbb{Z}$. By Assignment 3 Exercise 4, $\forall \epsilon > 0 \; \exists m \in E : \lfloor x \rfloor - \epsilon < m \leq \lfloor x \rfloor$; however, $\mathbb{Z}$ is discrete, which implies $\lfloor x \rfloor \in E$. Then $\lfloor x \rfloor \in \mathbb{Z}$ and $\lceil x \rceil \in \mathbb{Z}$. We want to prove that x is an interior point of $\mathbb{R} \setminus \mathbb{Z}$. Thus,

$$\lfloor x \rfloor < x < \lceil x \rceil \implies \epsilon_0 = \min\left\{\frac{x - \lfloor x \rfloor}{2}, \frac{\lceil x \rceil - x}{2}\right\} > 0 \implies \lfloor x \rfloor < x \pm \epsilon_0 < \lceil x \rceil;$$

hence, $\mathbb{R} \setminus \mathbb{Z}$ is open, as needed.

(c) No, the subset of rationals $\mathbb{Q}$ cannot be closed in the set of reals $\mathbb{R}$. We need to prove that $\mathbb{R} \setminus \mathbb{Q}$ is not open. Let $x \in \mathbb{R} \setminus \mathbb{Q}$. By the Density of $\mathbb{Q}$, $\forall \epsilon > 0 \; \exists r \in \mathbb{Q}$ such that $x - \epsilon < r < x + \epsilon$; hence, x is not an interior point of $\mathbb{R} \setminus \mathbb{Q}$. Thus, $\mathbb{R} \setminus \mathbb{Q}$ is not open, as needed.

(d) Let Λ be any index set and $F_\lambda \subset \mathbb{R}$ for each $\lambda \in \Lambda$. Assume F_λ is closed for all $\lambda \in \Lambda$. Then F_λ^c is open for all $\lambda \in \Lambda$. By Assignment 3 Exercise 5,

$$\bigcup_{\lambda \in \Lambda} F_\lambda^c = \{x \in \mathbb{R} : \exists \lambda \in \Lambda \text{ such that } x \in F_\lambda\}$$

is open. Thus,

$$\bigcap_{\lambda \in \Lambda} F_\lambda = \{x \in \mathbb{R} : x \in F_\lambda \text{ for all } \lambda \in \Lambda\} = \mathbb{R} \setminus \bigcup_{\lambda \in \Lambda} F_\lambda^c$$

is closed, as needed.

(e) Let $n \in \mathbb{N}$ and $F_1, ..., F_n \subset \mathbb{R}$. Assume $F_1, ..., F_n$ are closed. Then $F_1^c, ..., F_n^c$ are open. By Assignment 5 Exercise 5,

$$\bigcap_{m=1}^{n} F_m^c = \{x \in \mathbb{R} : x \in F_m \text{ for all } m = 1, ..., n\}$$

is open. Thus,

$$\bigcup_{m=1}^{n} F_m = \{x \in \mathbb{R} : x \in F_m \text{ for some } m = 1, ..., n\} = \mathbb{R} \setminus \bigcap_{m=1}^{n} F_m^c$$

is closed, as needed.

□

Exercise 2. Let $F \subset \mathbb{R}$ be a closed set, and let $\{x_n\}$ be a sequence of elements of F converging to $x \in \mathbb{R}$. Prove that $x \in F$. *Hint*: Assume that $x \in F^c$ and arrive at a contradiction.

Proof. Let $F \subset \mathbb{R}$ be a nonempty set. Suppose F is closed and $\{x_n\}$ is any sequence of F converging to $x \in \mathbb{R}$. Then F^c is open and $\forall \epsilon > 0\ \exists M \in \mathbb{N}\ \forall n \geq M : |x_n - x| < \epsilon$. Assume, for the purpose of contradiction, $x \in F^c$. Then $\exists \epsilon_0 > 0 : (x - \epsilon_0, x + \epsilon_0) \subset F^c$ and hence $\forall n \in \mathbb{N} : |x_n - x| \geq \epsilon_0$. Thus, $\exists \epsilon_0 > 0\ \forall M \in \mathbb{N}\ \exists n_0 \geq M : |x_n - x| \geq \epsilon$ and $x_n \not\to x$, which is a contradiction. Indeed, $x \in F$, as wanted. □

Exercise 3. Prove that if $\{x_n\}$ is a convergent sequence, $k \in \mathbb{N}$, then

$$\lim_{n\to\infty} x_n^k = \left(\lim_{n\to\infty} x_n\right)^k.$$

Hint: Use induction.

Proof. Let $\{x_n\}$ be any convergent sequence of $\mathbb{R}$. We want to inductively prove that

$$\forall k \in \mathbb{N} : \lim_{n\to\infty} x_n^k = \left(\lim_{n\to\infty} x_n\right)^k.$$

1) (Base Case)

$$\lim_{n\to\infty} x_n^1 = \lim_{n\to\infty} x_n = \left(\lim_{n\to\infty} x_n\right)^1.$$

2) (Inductive Step)

$$\begin{aligned}\lim_{n\to\infty} x_n^m = \left(\lim_{n\to\infty} x_n\right)^m \implies \lim_{n\to\infty} x_n^{m+1} &= \left(\lim_{n\to\infty} x_n^m\right)\left(\lim_{n\to\infty} x_n\right) \\ &= \left(\lim_{n\to\infty} x_n\right)^m \left(\lim_{n\to\infty} x_n\right) = \left(\lim_{n\to\infty} x_n\right)^{m+1}.\end{aligned}$$

By the Principle of Induction,

$$\forall k \in \mathbb{N} : \lim_{n\to\infty} x_n^k = \left(\lim_{n\to\infty} x_n\right)^k,$$

as wanted. □

Exercise 4. Let

$$x_n := \frac{n - \cos n}{n}.$$

Use the Squeeze Theorem to show that $\{x_n\}$ converges and find the limit.

Proof. Let
$$x_n := \frac{n - \cos n}{n} = 1 - \frac{\cos n}{n} \in \mathbb{R}.$$
We want to show that $\{x_n\}$ is bounded. By the Unit Property of Cosine Functions,
$$\forall n \in \mathbb{N} : -1 \le \cos n \le 1 \implies -\frac{1}{n} \le \frac{\cos n}{n} \le \frac{1}{n} \implies 1 - \frac{1}{n} \le 1 - \frac{\cos n}{n} \le 1 + \frac{1}{n}.$$
Then
$$\forall n \in \mathbb{N} : 1 - \frac{1}{n} \le x_n \le 1 + \frac{1}{n}.$$
We want to show that $\{x_n\}$ is convergent. By the Squeeze Theorem of Real Functions,
$$1 \leftarrow 1 - \frac{1}{n} \le x_n \le 1 + \frac{1}{n} \to 1 \implies \lim_{n\to\infty} x_n = 1.$$
Thus, $x_n \to 1$ as $n \to \infty$, as desired. □

Exercise 5. Let $S \subset \mathbb{R}$ be bounded above, and let x_0 be an upper bound for S. Prove that $x_0 = \sup S$ if and only if there exists a sequence $\{x_n\}$ of elements of S such that
$$\lim_{n\to\infty} x_n = x_0.$$
Hint: By Assignment 3, if $x_0 = \sup S$ then for all $n \in \mathbb{N}$ there exists $x_n \in S$ such that
$$x_0 - \frac{1}{n} < x_n \le x_0.$$

Proof. Let $S \subset \mathbb{R}$ be any bounded above set. Assume x_0 is an upper bound for S. We want to prove that
$$x_0 = \sup S \implies \exists \text{ sequence } \{x_n\} \text{ of } S : \lim_{n\to\infty} x_n = x_0.$$
If $x_0 = \sup S$, then
$$\forall \epsilon > 0\ \exists x \in S : x_0 - \epsilon < x \le x_0 \implies \forall n \in \mathbb{N}\ \exists x_n \in S : x_0 - \frac{1}{n} < x_n < x_0 + \frac{1}{n}$$
and
$$\forall n \in \mathbb{N} : 0 \le |x_n - x_0| < \frac{1}{n} \implies \lim_{n\to\infty} |x_n - x_0| = 0 \implies \lim_{n\to\infty} x_n = x_0.$$
We want to conversely prove that
$$\exists \text{ sequence } \{x_n\} \text{ of } S : \lim_{n\to\infty} x_n = x_0 \implies x_0 = \sup S.$$
If $\exists$ sequence $\{x_n\}$ of S such that $x_n \to x_0$, then
$$\forall \epsilon > 0\ \exists n \in \mathbb{N} : |x_n - x_0| < \epsilon \implies \forall \epsilon > 0\ \exists x_n \in S : x_0 - \epsilon < x_n < x_0 + \epsilon$$
and
$$\forall \epsilon > 0\ \exists x \in S : x_0 - \epsilon < x \le x_0 \text{ (because } x_0 \text{ is an upper bound for } S) \implies x_0 = \sup S.$$
□

Exercise 6. Let $E \subset \mathbb{R}$ be a nonempty set of real numbers. We say $x \in \mathbb{R}$ is a cluster point of E if for every $\epsilon > 0$
$$(x - \epsilon, x + \epsilon) \cap E \setminus \{x\} \neq \emptyset.$$
Said less formally, x is a cluster point of E if every interval containing x contains at least one element of E other than x.

(a) Prove that x is a cluster point of E if and only if there exists a sequence $\{x_n\}$ of elements of $E \setminus \{x\}$ such that

$$\lim_{n\to\infty} x_n = x.$$

Hint: If x is a cluster point of E, then for all $n \in \mathbb{N}$ there exists $x_n \in E$ with $x_n \neq x$ such that

$$x - \frac{1}{n} < x_n < x + \frac{1}{n}.$$

(b) Prove that the set of all cluster points of E is closed.

Proof. Let $E \subset \mathbb{R}$ be any nonempty set.

(a) We want to prove that x is a cluster point of E if and only if $\exists$ sequence $\{x_n\}$ of $E \setminus \{x\}$ such that $x_n \to x$. Assume, for proof, $x \in \mathbb{R}$. If x is a cluster point of E, then

$$\forall \epsilon > 0 : (x - \epsilon, x + \epsilon) \cap E \setminus \{x\} \neq \emptyset \implies \forall n \in \mathbb{N}\ \exists x_n \in E : 0 < |x_n - x| < \frac{1}{n}$$

and hence

$$\lim_{n\to\infty} |x_n - x| = 0 \implies \lim_{n\to\infty} x_n = x.$$

If conversely $\exists$ sequence $\{x_n\}$ of $E \setminus \{x\}$ such that $x_n \to x$, then

$$\forall \epsilon > 0\ \exists n \in \mathbb{N} : 0 < |x_n - x| < \epsilon \implies \forall \epsilon > 0\ \exists x_n \in E : x_n \in (x - \epsilon, x) \cup (x, x + \epsilon)$$

and hence

$$\forall \epsilon > 0 : (x - \epsilon, x + \epsilon) \cap E \setminus \{x\} \neq \emptyset \implies x \text{ is a cluster point of } E.$$

(b) We want to prove that the set of all cluster points of E is closed. Assume, for proof, x is not a cluster point of E. Then

$$\exists \epsilon_0 > 0 : (x - \epsilon_0, x + \epsilon_0) \cap E \setminus \{x\} = \emptyset \implies (x - \epsilon_0, x) \cup (x, x + \epsilon_0) \subset E^c$$

and

$$(x - \epsilon_0, x) \cup (x, x + \epsilon_0) \text{ is open} \implies (x - \epsilon_0, x + \epsilon_0) \text{ is not a cluster interval of } E.$$

$\square$